5.6 Graphing Linear Inequalities in Two Variables

Name _____ Period ____ Date _____

Algebra I 5.6 (30-minute plan) · Student sheet

I can…

- check solutions of linear inequalities in two variables.
- graph linear inequalities in two variables.
- write and use linear inequalities for real-life problems.

Vocabulary · fill in the missing word

Linear inequality in two variables — _____

Solution — an ordered pair that makes the _____ true

Boundary line — _____ for < or >, solid for ≤ or ≥

Half-plane — the side of the boundary line that contains the _____

Part 1 · Notes and examples (fill in each note, then work the example)

Example 1

Is (2, −3) a solution of $3x - 2y \geq 12$?

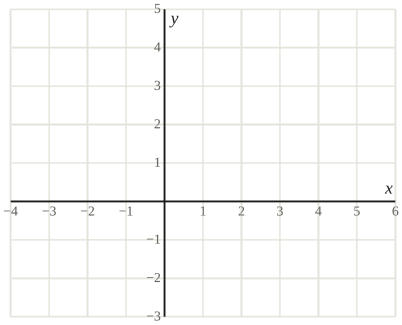
Boundary line: Boundary line: _____ for < or >, _____ for ≤ or ≥.

Test point: Test a point not on the line. If it is true, shade _____ side.

Shading: y > mx + b: shade _____; y < mx + b: shade _____.

Example 2

Graph $y \leq -\frac{1}{2}x + 2$.



Part 2 · Partner practice

Partner practice

Is $(-1, 4)$ a solution of $y < -3x + 2$?